

Good and bad kernels

I discuss which covariance kernels one should use and which to avoid when constructing a design with a Gaussian random fields. If the kernel is real analytic (e.g., the Gaussian kernel), the MSE tends to zero as the number of design points increases regardless of where these points are placed. For such kernels it is difficult to distinguish a good design from a bad one. If the kernel is rougher, the design must be space-filling if the MSE is to tend to zero.